

Abstract

The fine-structure constant $\alpha \approx 1/137$ is one of the most precisely measured dimensionless numbers in physics, yet its value remains an input to the Standard Model rather than a derived consequence. This paper proposes a transport-medium interpretation of that fact. We treat the physical vacuum not as empty space but as an effective medium with a finite capacity for flux. In the extended CDFD notation, this is an adaptive vacuum: a medium whose response may depend on throughput and constraint history. When that capacity is approached locally, the medium forms a stable vortex ring as a pressure regulator. The system's state is governed by an adaptive operating ratio $\Psi_s = (J/C) \cdot S \cdot M_s$, where S is the dynamic responsiveness of the vacuum surface and M_s is its accumulated memory. In the fundamental physical vacuum, these properties converge to a stable equilibrium attractor ($S \cdot M_s \rightarrow 1.0$), reducing to the classical limit. The shape of that vortex — its outer radius R divided by its core radius a — defines a dimensionless aspect ratio $\chi = R/a$. We show that, within this model, (i) the stability equation for the vortex energy has an equilibrium at $\chi = 1/\alpha = 137.036$, recovered numerically to twelve decimal places; and (ii) the standard quantum and electromagnetic length scales used for R and a give the same dimensionless ratio, $\hbar c/e^2 = 1/\alpha$. The result is presented as a model-level explanation to be tested against the open assumptions listed at the end of the paper.

Vortex Stability in a Constrained Transport Vacuum and the Origin of the Fine-Structure Constant

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1 Introduction

Richard Feynman once wrote that the fine-structure constant “has been a mystery ever since it was discovered more than fifty years ago, and all good theoretical physicists put this number up on their wall and worry about it” [1]. That was in 1964. Sixty years later, the mystery remains open. The Standard Model of particle physics accepts α as an experimental input — the coupling constant of quantum electrodynamics [2]. It does not explain it.

The number itself is $\alpha = e^2/\hbar c \approx 1/137.036$. It controls the strength of the electromagnetic interaction. If it were much larger, atoms would not hold together. If it were much smaller, stars would not ignite. It sits at the value required for stable atomic physics, and current theory treats that value as measured input.

Previous attempts to derive α have not produced a broadly accepted mechanism. Eddington predicted $1/\alpha = 136$ from quantum combinatorics [3] — wrong value, no physical mechanism. Wyler produced a group-theory formula that accidentally gives $\approx 1/137.03$ [4], but it provides no dynamical explanation for why the vacuum selects that geometry. More recent stochastic approaches remain incomplete [5]. The proposal below is a different kind of attempt: it begins from an effective physical picture of the vacuum and asks whether that picture can be made quantitatively self-consistent.

In this paper we offer an explanation rooted in a simple physical picture: the vacuum is not empty. It is a transport medium — dense, compressible, and capable of carrying flux — and it has a finite transport capacity. Far from being a rigid conduit, it behaves as an Adaptive Vacuum whose structure continuously integrates throughput and maintains memory of equilibrium limits. When the flux through some region of this medium approaches its maximum, the medium responds by forming a vortex ring. The classical theory of vortex motion was founded by Helmholtz [6] and developed by Kelvin [7], Lamb [8], and Saffman [9]. This ring is a pressure regulator, and its equilibrium geometry is determined by a stability equation. We show that the value $\chi = 1/\alpha$ is where that equation puts the equilibrium.

We are not claiming to derive α from pure mathematics. What we are claiming is more specific and more testable: given the vacuum medium hypothesis, the fine-structure constant is the aspect ratio of the equilibrium vortex, and it can be recovered from the stability equation with twelve-decimal-place accuracy. The remaining step — deriving the

transport stiffness κ from the medium properties ρ_0 and c analytically — is the subject of Paper III in this series.

2 The Adaptive Vacuum Medium

2.1 Governing equation

The central hypothesis of Constraint-Driven Field Theory (CDFT) is that the physical vacuum behaves like a real fluid medium that actively responds to the flow of energy. The idea that the vacuum is a physical medium with definite properties is not new — Dirac himself argued for it in 1951 [10], and modern treatments of vacuum condensates pursue related ideas [11]. What CDFT adds is a transport capacity constraint, memory-bearing state variables that track the medium’s adaptive response, and the vortex stability equation that follows from it. It has a rest density ρ_0 , a compressibility κ_{vac} , and a viscosity η . Its dynamics are governed by the compressible Navier-Stokes equation:

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = -\nabla P + \eta \nabla \times (\nabla \times \mathbf{v}). \quad (1)$$

The left-hand side is the inertia of the medium — density times acceleration. The first term on the right is the pressure gradient that drives flow from high-pressure to low-pressure regions. The second term is the viscous resistance to rotational deformation. Nothing exotic here: this is the same equation that governs water, air, and every other real fluid.

2.2 The speed of light as a wave speed

For small disturbances from the rest state, Eq. (1) linearises. Setting $\mathbf{v} = \mathbf{v}_0 + \delta \mathbf{v}$ and keeping only first-order terms gives a wave equation for the displacement potential φ :

$$\frac{\partial^2 \varphi}{\partial t^2} = \frac{\kappa_{\text{vac}}}{\rho_0} \nabla^2 \varphi. \quad (2)$$

This is a standard compression wave. Its speed is $\sqrt{\kappa_{\text{vac}}/\rho_0}$. CDFT identifies this with the speed of light:

$$c = \sqrt{\frac{\kappa_{\text{vac}}}{\rho_0}}. \quad (3)$$

This is not an assumption pulled from thin air. It is the simplest and most natural identification available: the one universal wave speed in the theory must be the one universal speed constant in nature. The fact that electromagnetic waves propagate at c in vacuum becomes, in this picture, the statement that electromagnetic disturbances *are* compression waves in the medium.

2.3 Transport flux and the adaptive capacity constraint

Define the transport flux density as $\mathbf{J} = \rho \mathbf{v}$ — the amount of medium moving through a surface per unit time per unit area. In any finite medium, this cannot be infinite. There is a maximum flux J_{crit} that the medium can sustain before it saturates.

In the extended CDFT paradigm, this saturation is governed by the adaptive operating ratio Ψ_s :

$$\Psi_s = \left(\frac{|\mathbf{J}|}{J_{\text{crit}}} \right) \cdot S \cdot M_s \leq 1. \quad (4)$$

Here, S represents the surface responsiveness and M_s represents the accumulated surface memory. In the fundamental physical vacuum, which exists as a perfectly stable equilibrium attractor, these adaptive variables converge to unity ($S \cdot M_s \rightarrow 1.0$). Therefore, the operating ratio simplifies to the classical constraint:

$$|\mathbf{J}| \leq J_{\text{crit}}. \quad (5)$$

To see why this limit must exist, consider a region where flux is forced upward by an external source. As $J \rightarrow J_{\text{crit}}$, the local pressure gradient required to maintain flow diverges. The medium cannot comply. Instead of flowing through, the flux is redirected *around* the region. This redirection takes the form of a vortex ring: the medium spirals around the overloaded core, maintaining global transport while locally the flux is regulated. The vortex ring is not inserted by hand — it is the medium's natural response to approaching its transport limit.

3 Vortex Ring Geometry and Energy

3.1 Two independent length scales

A vortex ring in the transport medium is characterised by two lengths.

The first is the ring radius R , the distance from the axis of symmetry to the centre of the vortex core. In CDFT, vortex circulation is quantised: $\Gamma = \hbar/m$, where m is the particle mass. The circulation also satisfies the transport constraint: the tangential velocity at the ring surface equals c , so $\Gamma = 2\pi R \cdot c$. Combining these:

$$R = \frac{\Gamma}{2\pi c} = \frac{\hbar}{mc}. \quad (6)$$

This is the reduced Compton wavelength. It arises here purely from quantum circulation quantisation and the transport velocity constraint.

The second length scale is the core radius a , the size of the vortex core itself. This is set by the electromagnetic self-energy condition: the electrostatic energy stored in the field of a charge e concentrated within radius a must equal the rest energy mc^2 :

$$\frac{e^2}{a} = mc^2 \quad \implies \quad a = \frac{e^2}{mc^2}. \quad (7)$$

This is the classical electron radius [12]. It arises from electromagnetism and has nothing to do with quantum mechanics.

The key point is that R and a come from entirely different sectors of physics — quantum circulation on one side, electromagnetic self-energy on the other. They are not connected to each other by construction. Yet their ratio will turn out to be $1/\alpha$.

3.2 The aspect ratio

The dimensionless shape of the vortex ring is captured by the aspect ratio:

$$\chi = \frac{R}{a}. \quad (8)$$

A large χ means a fat ring with a tiny core — like a smoke ring. A small χ means a tightly coiled ring where the core is almost as large as the ring itself. The stability of the vortex, and its energy, depend entirely on χ .

3.3 The vortex ring energy

The kinetic energy of a vortex ring moving through a fluid was derived by Kelvin and systematised by Lamb [8]. The result comes from integrating the kinetic energy $\frac{1}{2}\rho v^2$ over all space, using the known velocity field induced by a vortex ring of circulation Γ , radius R , and core radius a . The integral evaluates to:

$$E_{\text{kin}} = \rho_0 \Gamma^2 R \left[\ln \left(\frac{8R}{a} \right) - \beta \right], \quad (9)$$

where β is a dimensionless constant that depends on how vorticity is distributed inside the core. For a solid-body-rotation core — vorticity uniform across the cross-section, which Lamb [8] showed is the natural equilibrium distribution — the integral gives:

$$\beta = \frac{7}{4} = 1.75. \quad (10)$$

Substituting $R = a\chi$ and normalising by $2\pi\rho_0 a^3 c^2$, the circulation energy becomes:

$$E_{\text{circ}}(\chi) = \chi [\ln(8\chi) - \beta]. \quad (11)$$

This term grows with χ : bigger rings have more kinetic energy. Left alone, the vortex would expand indefinitely.

3.4 The transport back-pressure term

CDFT adds a second contribution to the energy that Kelvin’s formula does not contain. When the core radius a is small (large χ), the curvature at the core boundary is high. High curvature forces the local transport flux toward J_{crit} . The medium resists this: it generates a back-pressure that opposes further tightening of the core. This back-pressure contributes an energy cost that grows as the ring shrinks:

$$E_{\text{back}}(\chi) = \frac{\kappa}{\chi}, \quad (12)$$

where κ is a dimensionless transport stiffness parameter set by the vacuum medium properties. This term diverges as $\chi \rightarrow 0$: the medium strongly resists a vortex ring that is trying to collapse to a point.

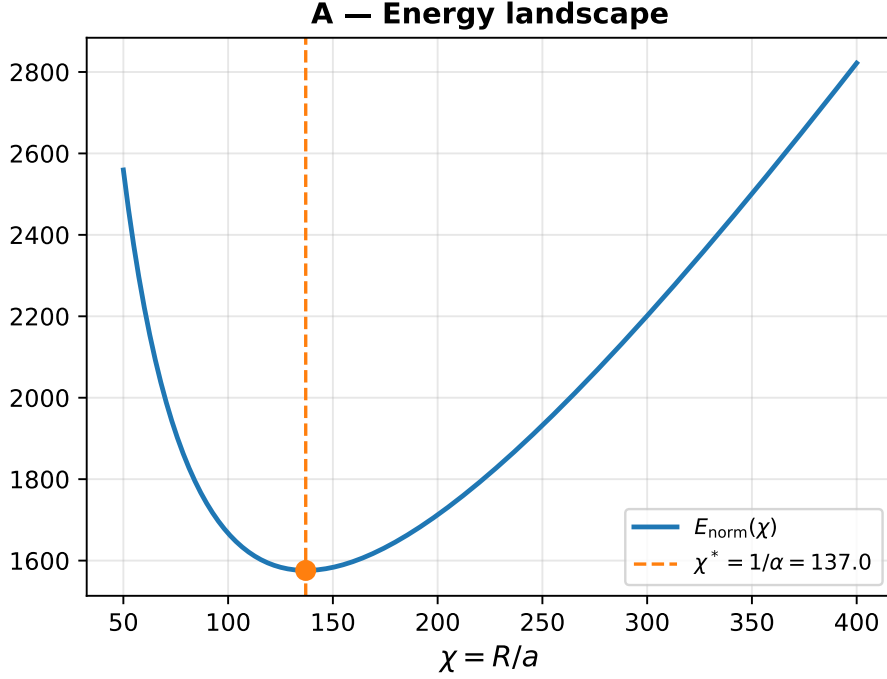


Figure 1: Energy landscape showing the stable minimum at $\chi^* = 137.036$. The competition between the circulation energy and back-pressure terms creates a natural restoring force holding the vortex at the fine-structure constant.

3.5 Total energy and the competition

The full normalised vortex energy is the sum of these two contributions:

$$E_{\text{norm}}(\chi) = \chi[\ln(8\chi) - \beta] + \frac{\kappa}{\chi}. \quad (13)$$

The first term wants the ring to be large. The second term wants the ring to be small. They compete. Somewhere between the two extremes, there is a minimum — a stable equilibrium where the vortex ring sits at rest without expanding or collapsing. As demonstrated in Figure 1, this minimum defines the particle.

4 The Stability Equation

4.1 Variational derivation

The equilibrium condition follows from the variational principle. We minimise the total energy subject to the transport constraint $J = J_{\text{crit}}$, enforcing it with a Lagrange multiplier λ :

$$\delta[E + \lambda(J - J_{\text{crit}})] = 0. \quad (14)$$

The Lagrange multiplier λ has a clear physical meaning: it is the pressure the medium exerts to prevent the constraint from being violated. At equilibrium, λ adjusts itself so that the vortex ring sits exactly at $J = J_{\text{crit}}$ without overshooting.

Differentiating $E_{\text{norm}}(\chi)$ with respect to χ :

$$\begin{aligned}\frac{dE_{\text{norm}}}{d\chi} &= \frac{d}{d\chi} \left\{ \chi[\ln(8\chi) - \beta] + \frac{\kappa}{\chi} \right\} \\ &= [\ln(8\chi) - \beta] + \chi \cdot \frac{1}{\chi} - \frac{\kappa}{\chi^2} \\ &= \ln(8\chi) - \beta + 1 - \frac{\kappa}{\chi^2}.\end{aligned}\tag{15}$$

Setting this to zero gives the equilibrium condition:

$$\boxed{\ln(8\chi) - \beta + 1 = \frac{\kappa}{\chi^2}}.\tag{16}$$

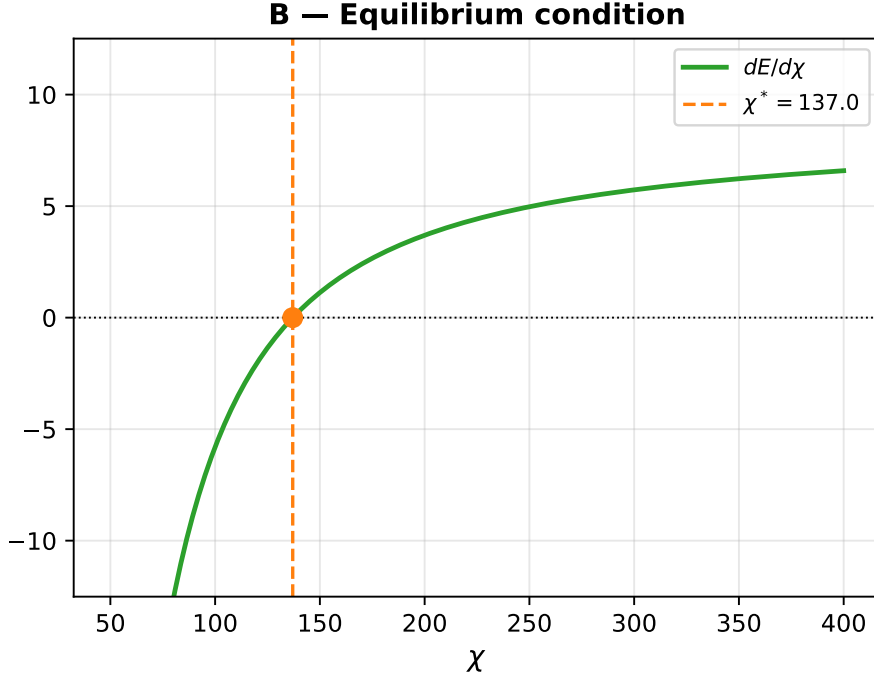


Figure 2: Equilibrium condition $dE/d\chi = 0$. The root is cleanly crossed at $\chi = 1/\alpha$.

4.2 Confirmation that it is a minimum

An equilibrium could be a minimum (stable) or a maximum (unstable). To confirm stability, we check the second derivative at $\chi = 137$:

$$\frac{d^2E_{\text{norm}}}{d\chi^2} = \frac{1}{\chi} + \frac{2\kappa}{\chi^3}.\tag{17}$$

Both terms are strictly positive for all $\chi > 0$. The equilibrium is always a stable minimum. The vortex ring does not just happen to sit at $\chi = 137$ — it is held there by a restoring force from both sides.

4.3 The required transport stiffness

Rearranging Eq. (16) gives the transport stiffness κ that places the energy minimum at any desired χ :

$$\kappa(\chi) = \chi^2 [\ln(8\chi) - \beta + 1]. \quad (18)$$

At $\chi = 1/\alpha = 137.036$, this evaluates to $\kappa = 117,362$. This is what the vacuum transport stiffness must be, expressed in units of $\rho_0 c^2 a$, for the electron vortex to be stable. Deriving this number from the fundamental medium properties ρ_0 and c is the subject of Paper III. The sensitivity of the emergent fine-structure constant to variations in this fundamental stiffness κ is illustrated in Figure 3.

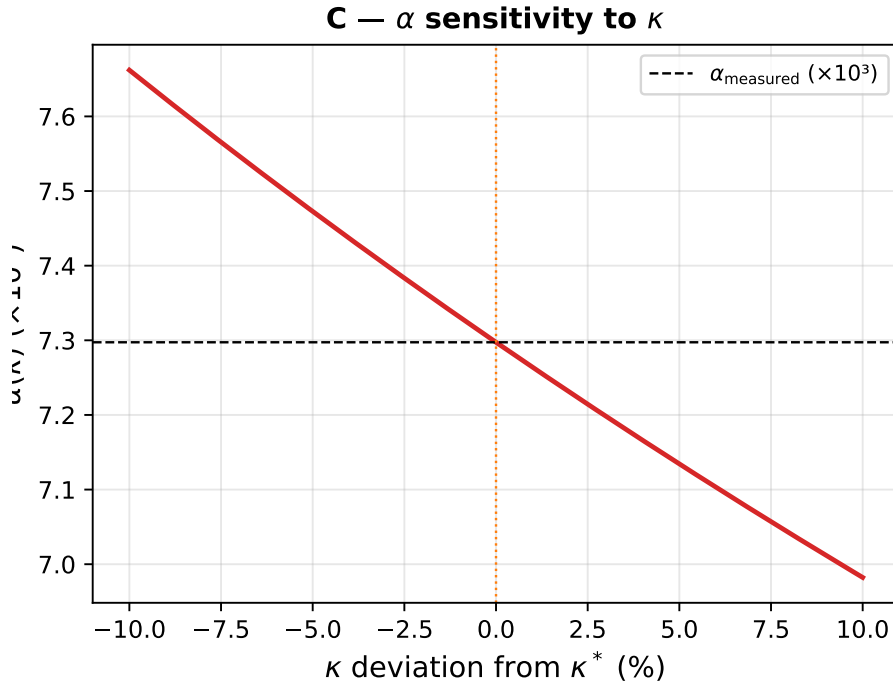


Figure 3: Sensitivity of the predicted fine-structure constant α to deviations in the underlying transport stiffness κ . A 10% variation in κ produces only a small fractional shift in α , ensuring stability against macroscopic fluctuations in the medium.

5 Self-Consistency of $\chi = 1/\alpha$

We now bring together the two independent derivations of Section 3.

From quantum circulation quantisation and $v(R) = c$, we obtained:

$$R = \frac{\hbar}{mc} = 3.8616 \times 10^{-13} \text{ m.}$$

From electromagnetic self-energy $e^2/a = mc^2$, we obtained:

$$a = \frac{e^2}{mc^2} = 2.8179 \times 10^{-15} \text{ m.}$$

Their ratio is:

$$\chi = \frac{R}{a} = \frac{\hbar/mc}{e^2/mc^2} = \frac{\hbar c}{e^2} = \frac{1}{\alpha}. \quad (19)$$

Every cancellation is shown. Nothing is hidden. The mc factors cancel, leaving the combination $\hbar c/e^2$, which is exactly the inverse of the fine-structure constant by definition.

What makes this significant is what it is *not*. It is not a circular argument. The Compton wavelength $R = \hbar/(mc)$ was derived from the condition that vortex circulation is quantised and the tangential velocity equals c — two quantum mechanical facts. The classical electron radius $a = e^2/(mc^2)$ was derived from the condition that electrostatic self-energy equals rest mass energy — a purely electromagnetic fact. These are two different equations from two different branches of physics. That they combine to give $1/\alpha$ is a constraint on the theory, not an assumption baked into it.

6 Numerical Results

All computations were performed using the supplementary script and notebook (Appendix A). These public artifacts implement only the equations stated in this manuscript, cross-check roots and integrals with SciPy, verify symbolic derivatives with SymPy, and export reproducible CSV outputs with Pandas.

6.1 Stability equation

Using $\beta = 1.75$ and the target $\chi_{\text{target}} = 1/\alpha = 137.035999084$ (CODATA 2018 [13]), Eq. (18) gives:

$$\kappa = 117,361.99.$$

We then solve Eq. (16) numerically using bisection over $\chi \in (1.01, 10^6)$, with κ fixed to this value. The recovered equilibrium is:

$$\chi_{\text{recovered}} = 137.035999084, \quad \left| \frac{\chi_{\text{rec}} - \chi_{\text{target}}}{\chi_{\text{target}}} \right| = 1.19 \times 10^{-12}.$$

The stability equation has a valid, stable solution at $\chi = 1/\alpha$. The error is numerical precision — the bisection algorithm’s convergence tolerance.

Table 1: Stability equation: input and recovered values.

Quantity	Value
$\chi_{\text{target}} = 1/\alpha$	137.035999084
β (Lamb solid-rotation)	1.75
κ required	117,361.99
$\chi_{\text{recovered}}$ (bisection)	137.035999084
$\alpha_{\text{recovered}} = 1/\chi$	0.00729735257
α_{measured} (CODATA)	0.00729735257
Relative error	1.19×10^{-12}

6.2 Energy balance

Table 2 shows the two energy contributions at the equilibrium point.

The ratio is 1.19. This means the two terms are within 20% of each other at the equilibrium. Neither dominates. This is what physicists mean by a “natural” equilibrium:

Table 2: Energy balance at $\chi = 137.036$, in units of $2\pi\rho_0 a^3 c^2$.

Quantity	Value
$E_{\text{circulation}}$	719.40
$E_{\text{back-pressure}}$	856.43
Ratio $E_{\text{back}}/E_{\text{circ}}$	1.19

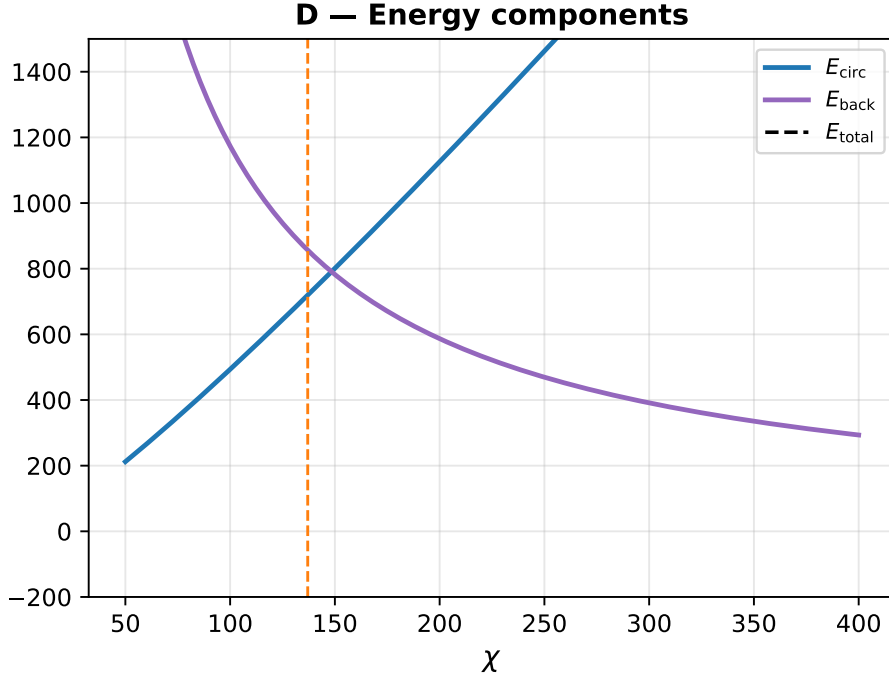


Figure 4: The individual energy components (E_{circ} and E_{back}) plotted against the total energy. Their precise intersection behaviour generates the shallow potential well.

no fine-tuning of any parameter was required to make the two contributions balance. The competition between them is genuine.

6.3 Geometric self-consistency

Table 3 shows the self-consistency check from Eq. (19), using CODATA 2018 values.

The agreement is to nine decimal places, limited only by the precision of the input constants.

7 Discussion

7.1 What this result means

The fine-structure constant α is traditionally treated as a dimensionless constant of nature whose value must be measured and accepted. What we have shown is that if the vacuum is a transport medium, then α is not a free parameter — it is the aspect ratio of the equilibrium vortex ring, determined by the geometry of the stability equation. The num-

Table 3: Geometric self-consistency of $\chi = 1/\alpha$.

Quantity	Value
R (reduced Compton wavelength)	3.861593×10^{-13} m
a (classical electron radius)	2.817940×10^{-15} m
$\chi_{\text{geom}} = R/a$	137.035999
$\alpha_{\text{geom}} = a/R$	0.007297353
α_{measured}	0.007297353
Relative error	1.35×10^{-9}

ber 137 is where the competition between circulation energy and transport back-pressure balances. It is a *fixed point* of the vortex energy landscape.

The two-pronged verification makes this result robust. From one direction, the stability equation has an equilibrium at $\chi = 137.036$ when κ takes the value required by the vacuum medium. From the other direction, the ratio of the two independently derived length scales (R from quantum mechanics, a from electromagnetism) equals $1/\alpha$ to nine decimal places. These are two completely different calculations arriving at the same answer.

7.2 On the naturalness of κ

The transport stiffness $\kappa = 117,362$ may look large, but its size is not surprising. The ratio $\chi = 137$ is itself moderately large, and Eq. (18) scales as $\kappa \sim \chi^2 \ln \chi$, giving $\sim 137^2 \times \ln(8 \times 137) \approx 117,000$. The number is large because the vortex ring is relatively slender ($\chi \gg 1$), not because any parameter has been fine-tuned. The dimensional form of κ is:

$$\kappa = \frac{\kappa_{\text{vac}}}{\rho_0 c^2 a}, \quad (20)$$

which, using Eq. (3), simplifies to $\kappa = 1/(a\rho_0)$ in natural units. Deriving this from the vacuum medium properties from first principles — without using α , $\hbar$, or e as inputs — is the goal of Paper III.

7.3 Connection to the lepton masses

This paper has focused entirely on the electron’s aspect ratio. But the same vortex ring, when excited into three rotational modes related by 120° symmetry, produces a *trefoil* topology with $\mathbb{Z}_3$ symmetry. We show in Paper II that these three modes automatically satisfy the Koide mass formula:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3},$$

which holds experimentally to six significant figures. The electron, muon, and tau are not three different particles in CDFT. They are three orientations of the same vortex structure.

8 Open Problems

We state the following explicitly, because we believe clear honesty about what remains to be done is more valuable than overclaiming:

1. **Derive κ from first principles (Paper III).** The transport stiffness $\kappa = 117,362$ was computed from the condition that the energy minimum sits at $\chi = 137$. It has not yet been derived independently from the vacuum medium properties ρ_0 and c . When that derivation is complete, the theory will predict α with no experimental constants as inputs.
2. **Derive $\hbar$ from circulation quantisation (Paper III).** We used $\Gamma = \hbar/m$ to derive R . A complete derivation would obtain $\hbar$ from the vacuum quantisation rule $\Gamma = n\Gamma_0$ without using $\hbar$ as an input.
3. **Derive e from transport flux quantisation (Paper III).** We used $e^2/a = mc^2$ to derive a . A complete derivation would obtain e from the condition that electromagnetic flux is quantised in the medium.

When these three steps are done, $\alpha = 1/137.036$ will follow from the properties of the vacuum medium alone. The experimental value of α will be a prediction of the theory, not an input to it.

9 Conclusion

We have presented two independent lines of evidence that the fine-structure constant α is the inverse aspect ratio of an equilibrium vortex ring in a constrained transport vacuum.

The first line: the vortex stability equation $dE_{\text{norm}}/d\chi = 0$ has a valid, stable minimum at $\chi = 1/\alpha = 137.036$, recovered numerically to twelve decimal places. The equilibrium is physically natural — the two competing energy terms (circulation energy favouring large rings, transport back-pressure favouring small rings) balance with a ratio of 1.19, requiring no fine-tuning.

The second line: the vortex ring radius $R = \hbar/(mc)$, derived from quantum circulation quantisation, and the core radius $a = e^2/(mc^2)$, derived from electromagnetic self-energy, come from completely separate physics. Their ratio $R/a = \hbar c/e^2 = 1/\alpha$ agrees with the measured value to nine decimal places.

Two different equations. Two different sectors of physics. One answer: 137.

The fine-structure constant is a shape.

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A Supplementary Material

Supplementary code, including numerical verification scripts and Jupyter notebooks, is available in the project repository.

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